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|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|           |                                                                                                                                                                                                                          |           |
| 7(a)(i)   | no current enters/leaves the input                                                                                                                                                                                       | <b>B1</b> |
| 7(a)(ii)  | gain is the same for all frequencies                                                                                                                                                                                     | <b>B1</b> |
| 7(b)(i)   | $V_{IN} = 1.5 \times 400 / (400 + 1100) = 0.40 \text{ V}$<br><b>or</b><br>$V_{IN} = 1.5 - (1.5 \times 1100 / 1500) = 0.40 \text{ V}$<br><b>or</b><br>$(1.5 - V_{IN}) / 1100 = V_{IN} / 400$ so $V_{IN} = 0.40 \text{ V}$ | <b>A1</b> |
| 7(b)(ii)  | gain = $(-) R_f / R_i$                                                                                                                                                                                                   | <b>C1</b> |
|           | $V_{OUT} / 0.40 = (360 + 100) / 96$                                                                                                                                                                                      | <b>C1</b> |
|           | $V_{OUT} = 1.9 \text{ V}$                                                                                                                                                                                                | <b>A1</b> |
| 7(b)(iii) | resistance <u>of thermistor</u> decreases                                                                                                                                                                                | <b>B1</b> |
|           | (magnitude of) gain decreases so reading decreases                                                                                                                                                                       | <b>B1</b> |
| 7(b)(iv)  | (at gain 12.5) $V_{OUT}$ is 5.0 V, so (above gain 12.5) output becomes saturated                                                                                                                                         | <b>B1</b> |